

Four-cycle supersaturation at orders 19, 20, and 21

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Abstract

For a finite simple graph, let $h(n)$ be the minimum number of non-induced four-cycles among graphs on n vertices with $\text{ex}(n, C_4) + 1$ edges. Using the attributed small-order extremal thresholds and complete order-19 extremal catalogue of Afzaly and McKay, together with exhaustive base generation by nauty, we obtain $h(19) = 2$ and $h(20) = h(21) = 3$. The main step excludes every 20-vertex, 47-edge graph with at most two four-cycles. A vertex-deletion argument reduces this step to the catalogue and a complete rooted enumeration of 19-vertex, 41-edge C_4 -free graphs of minimum degree at least three. Exact three-cycle constructions give the matching upper bounds. The computation uses integer arithmetic and has a separate verification by attachment-block enumeration. The original uniform square-root lower-bound problem remains unresolved.

1 Statement and scope

All graphs are finite, simple, and undirected. A four-cycle is counted once per cyclic edge set; it need not be induced. In particular, K_4 contains three four-cycles. Write $e_n = \text{ex}(n, C_4)$ and

$$h(n) = \min\{t_4(G) : |V(G)| = n, |E(G)| = e_n + 1\}.$$

The Erdős–Simonovits problem asks whether there are constants $c > 0$ and n_0 such that $h(n) \geq c\sqrt{n}$ for every $n \geq n_0$; see the discussion and references in Qiao–Zhan [2] and He–Ma–Yang [3]. The present result concerns only three orders.

Theorem 1 (Exact finite computation). *Using the Afzaly–McKay catalogue thresholds and complete five-class extremal census at order 19 as external inputs, and the completeness of the stated nauty base enumerations, one has*

$$h(19) = 2, \quad h(20) = h(21) = 3.$$

Here $(e_{19}, e_{20}, e_{21}) = (42, 46, 50)$.

The catalogue [1] states that entries whose edge and class counts are not preceded by inequality signs give both the exact Turán number and all extremal graphs. Its rows $(n, e_n, \text{classes}) = (19, 42, 5)$, $(20, 46, 1)$, and $(21, 50, 1)$ have this form. We use only the order-19 graph census, and use the following smaller thresholds in the reductions:

s	0	1	2	3	4	5	6	7	8	9	10	11	12	13
e_s	0	0	1	3	4	6	7	9	11	13	16	18	21	24

The entries through order three are immediate; the other entries are attributed catalogue inputs. We do not regenerate the complete order-19 extremal census or reprove these thresholds.

Qiao and Zhan [2, Theorem 5] give $h(n) = 1$ for $6 \leq n \leq 11$ and $h(12) = h(13) = 2$. The public computational working report [4] claims $h(n) = 2$ for $14 \leq n \leq 18$ and discloses its AI assistance and software dependencies. We do not recertify that range here. He, Ma, and Yang [3, Theorem 1.6 and Corollary 1.7] obtain exact supersaturation results at large special orders; the displayed equality $h(q^2 + q + 1) = q - 1$ in their corollary assumes $q = 2^k$ with $k \geq 40$, so it does not supply the $q = 4$ case considered here. The substantive finite step in this note is the exclusion at $(n, m) = (20, 47)$. It requires a near-extremal census in addition to the extremal graphs. Finite Zarankiewicz constructions, such as Zeng [6], provide adjacent context for explicit finite forbidden-subgraph certificates; they are not inputs to this proof.

No equivalent finite conclusion was found in the specifically inspected sources above. This is a bounded comparison, not a certification of priority. No assertion for larger orders, or resolution of the uniform square-root problem, follows from Theorem 1.

2 Counting and the reduction to order 19

For vertices x, y , let $\text{codeg}_G(x, y) = |N_G(x) \cap N_G(y)|$. Two elementary identities used throughout are

$$t_4(G) = \frac{1}{2} \sum_{\{x,y\} \subseteq V(G)} \binom{\text{codeg}_G(x,y)}{2}, \quad (1)$$

$$t_4(G + v) - t_4(G) = \sum_{\{x,y\} \subseteq S} \text{codeg}_G(x,y), \quad N_{G+v}(v) = S. \quad (2)$$

For the first identity, each cycle has two pairs of opposite vertices. For the second, a cycle through the new vertex v is uniquely determined by its two neighbors x, y on the cycle and their common neighbor in G . Call S *cycle-preserving* if every pair in S has codegree zero. Thus adjoining v with neighborhood S creates no four-cycle exactly when S is cycle-preserving.

For a nonedge xy of a C_4 -free graph H , let $p_H(xy)$ be the number of simple paths of length three from x to y in H . Every four-cycle of $H + xy$ contains xy , so

$$t_4(H + xy) = p_H(xy). \quad (3)$$

In particular, $h(n) = 1$ if and only if an extremal C_4 -free graph has a nonedge with path count one: delete an edge of the unique cycle for the converse. This equivalence only decides the value one. For a graph with edge-disjoint cycles, one deletion cannot remove all cycles. The exact-one test and the neighborhood bound used below also appear in the working report [4]; we claim no originality for them.

Lemma 2. *Under the catalogue inputs, $h(19) = 2$. Moreover, every graph on 19 vertices with at least 44 edges has at least three four-cycles.*

Proof. The five order-19 catalogue records have 42 edges, are C_4 -free, and have minimum nonedge path counts

$$2, \quad 3, \quad 3, \quad 2, \quad 2, \quad (4)$$

in catalogue order. These values are computed over all $5 \binom{19}{2} - 5 \cdot 42 = 645$ nonedges. They exclude a one-cycle 43-edge graph by the exact-one test and catalogue completeness. Adding edge

$\{0, 18\}$ to the first catalogue record yields exactly two cycles, with vertex sequences $(0, 2, 16, 18)$ and $(0, 10, 15, 18)$, using its zero-based labels. This proves $h(19) = 2$.

For the second assertion, if necessary retain 44 edges of a hypothetical counterexample. There is a cycle since $44 > e_{19}$. Deleting one of its edges leaves 43 edges and at most one cycle, a contradiction. $\square$

Lemma 3. *If a graph G has 20 vertices, 47 edges, and at most two four-cycles, then it has a degree-four vertex v lying on no four-cycle. For $F = G - v$ and $S = N_G(v)$, the graph F has 19 vertices, 43 edges, and exactly two four-cycles, and S is a cycle-preserving four-set in F . Every vertex on a four-cycle of G has degree at least five.*

Proof. The average degree is $94/20 < 5$, so some vertex has degree at most four. No vertex can have degree at most three: deleting it would leave at least 44 edges on 19 vertices, contrary to Lemma 2. Deleting any degree-four vertex leaves 43 edges, which must retain at least two cycles. Consequently G has exactly two cycles, neither using that vertex. The same argument applies to every degree-four vertex, so all cycle vertices have degree at least five. Equation (2) proves the assertion about S . $\square$

There are two exhaustive cases for the cycles of F in Lemma 3. If they share an edge e , then $H = F - e$ is C_4 -free with 42 edges. It occurs in the catalogue, and $p_H(e) = 2$. Checking its path-count-two nonedges and then all four-sets therefore covers this case.

Suppose instead that the two cycles are edge-disjoint. There are edges e of the first and f of the second with disjoint endpoints. Indeed, otherwise take opposite edges ab, cd of the first cycle. Every edge of the second would meet both, hence lie between $\{a, b\}$ and $\{c, d\}$. The second cycle would then use all four edges of this $K_{2,2}$, including the other two edges of the first cycle, a contradiction.

Now $H = F - e - f$ is C_4 -free with 41 edges and

$$\delta(H) \geq 3, \quad \{x : d_H(x) = 3\} \subseteq S. \quad (5)$$

An endpoint of e or f lies on a cycle of G , has degree at least five in G , and loses at most the edge to v and one of e, f . Every other vertex has degree at least four and loses at most its edge to v . This proves the first assertion. Outside S , a deleted-edge endpoint has H -degree at least four and any other vertex also has H -degree at least four, proving the second. Each individual insertion restores precisely one of the two cycles, so $p_H(e) = p_H(f) = 1$. After both insertions one must check that no additional mixed cycle appears. Thus it suffices to enumerate all H satisfying (5), all such nonedge pairs, and all relevant four-sets.

3 A complete rooted census

Let r be a degree- d vertex of a C_4 -free graph H on n vertices. Put $A = N_H(r)$ and $B = V(H) \setminus (A \cup \{r\})$. The graph induced by A is a matching: a length-two path in A would give a four-cycle through r . Each vertex of B has at most one neighbor in A , for the same reason. Therefore

$$|E(H)| \leq n - 1 + \lfloor d/2 \rfloor + e_{n-1-d}. \quad (6)$$

At $(n, m) = (19, 41)$, substitution of the stated smaller thresholds gives an upper bound at most 39 for every $7 \leq d \leq 18$. Since $82/19 > 4$, the maximum degree of H is five or six. We root at a vertex of maximum degree $d \in \{5, 6\}$.

Let $b = |B| = 18 - d$, let k be the number of matching edges in A , and let t be the number of edges between A and B . Then

$$41 = d + k + t + |E(H[B])|, \quad 0 \leq k \leq \lfloor d/2 \rfloor, \quad 0 \leq t \leq b. \quad (7)$$

Because $\delta(H) \geq 3$ and a B -vertex has at most one neighbor outside B , the minimum degree in $H[B]$ is at least two. Its maximum degree is at most d .

For $d = 6$, the complete possibilities for $(|E(H[B])|, k, t)$ are

$$(20, 3, 12), \quad (21, 2, 12), \quad (21, 3, 11).$$

For $d = 5$, one has $21 \leq |E(H[B])| \leq 24$ and $k = 0, 1, 2$; the number of unattached vertices of B is

$$b - t = |E(H[B])| + k - 23.$$

Keep exactly the choices for which this number lies in $\{0, 1, 2, 3\}$. These are all solutions of (7) in the stated ranges.

The following nauty 2.9.3 commands [5] generate the bases:

```
geng -f -d2 -D5 13 21:24
geng -f -d2 -D6 12 20:21
```

Here `-f` excludes four-cycles and `-d/-D` bound the minimum/maximum degrees. There is no connectivity, bipartiteness, triangle-free, or residue-class restriction. The completed runs output 9,238 and 85 base graphs, respectively. Completeness of these two small enumerations is a software dependency of the theorem.

For each base, fix a matching of size k on the d labeled vertices of A . Give each attached vertex of B the color of its unique neighbor in A ; the other vertices are uncolored. The exact conditions on the coloring are:

1. Two vertices of B with the same color have no common neighbor in B .
2. No edge of B joins vertices whose colors are matched in A .
3. A matched color has between 1 and $d - 2$ vertices, and an unmatched color has between 2 and $d - 1$ vertices. Every B -vertex of degree two is colored, and a B -vertex of degree d is uncolored.

The first two conditions are equivalent to absence of mixed four-cycles. A cycle with one vertex of A and three of B violates the first; one using a matching edge of A violates the second. A cycle through r would require a length-two path in A or a twice-attached vertex of B . Two nonadjacent A -vertices on a cycle also require a twice-attached B -vertex. The remaining cycles lie within the already C_4 -free base. The third condition expresses the final minimum degree three and maximum degree d .

The production enumeration recursively assigns colors, using the permutations of A that preserve its matching. At each step it keeps one representative from each color orbit under the subgroup fixing every color already used. This does not change previous assignments and retains a representative of every coloring orbit. Remaining-degree and capacity pruning discard only assignments unable to satisfy the third condition. Each output is checked directly for its order, edge count, degree bounds, and C_4 -freeness.

d	$ B $	Base edge range	Base records	Rooted records
5	13	21–24	9,238	953
6	12	20–21	85	121

The 1,074 outputs cover every required graph up to isomorphism with a suitable root. They are rooted records, not 1,074 unrooted isomorphism classes. The lower-bound test does not require a further unrooted isomorphism reduction.

A separately implemented verifier enumerates feasible attachment *blocks*, pairs blocks corresponding to matched vertices of A , and solves the resulting disjoint exact-cover problems. It does not use the production color recursion or its stabilizers. Across 10,289 attachment cases on 9,323 bases, every case's complete set of attachment orbits agrees with the production records. Thus the comparison checks the sets of outputs, not just their cardinalities.

4 Exclusion and matching constructions

Proposition 4. *Under the stated catalogue and enumeration dependencies, every graph on 20 vertices with 47 edges contains at least three four-cycles.*

Proof. Apply Lemma 3. In the shared-edge branch, the five catalogue records give 25 path-count-two single-edge augmentations. For each, all $\binom{19}{4} = 3,876$ neighborhoods are checked by (2); none is cycle-preserving.

In the edge-disjoint branch, the rooted census covers $H = F - e - f$ by (5) and the preceding completeness argument. The separate verifier checks every pair of distinct path-count-one nonedges in all 1,074 records, including pairs with overlapping endpoints. It retains only augmentations with exactly two four-cycles and checks every four-element neighborhood. This larger search dispenses with the necessary degree-three-neighborhood filter and therefore includes every case needed in the reduction. The exact completed counts are:

	Shared-edge branch	Two-edge branch
Input records	5	1,074
Nonedges / nonedge pairs inspected	645	1,660
Two-cycle augmentations	25	674
Four-sets inspected	96,900	2,612,424
Cycle-preserving four-sets	0	0

In both branches every augmentation fails the necessary extension test. This excludes G . □

The computations in the proposition use exact adjacency sets and integer counts. The separate verifier decodes the graph records independently and also counts cycles by traversing them from their smallest vertex. All 645 catalogue nonedge counts were checked both by paths and by cycle traversals; the ten pairs of catalogue graphs were checked for nonisomorphism. The archive described below contains the programs, graph records, and full counters.

For the upper bounds, work over $\mathbb{F}_4 = \mathbb{F}_2[\alpha]/(\alpha^2 + \alpha + 1)$. Take the 21 projective points of $\mathbb{F}_4^3$, normalized so that the first nonzero coordinate is one, and join distinct points x, y when $x_1y_1 + x_2y_2 + x_3y_3 = 0$. For distinct projective points, their two orthogonality equations are independent and have a one-dimensional solution space. Hence they have at most one common projective neighbor, and the graph is C_4 -free.

For explicit labels, write the field elements as 0, 1, 2, 3, with $2 = \alpha$ and $3 = 1 + \alpha$, and list the normalized triples in lexicographic order, starting at label zero. The graph has 50 edges. Adding $\{2, 6\}$ gives exactly the three cycles

$$(1, 5, 2, 6), \quad (2, 6, 14, 15), \quad (2, 6, 18, 20).$$

For order 20, delete point 2, namely $(0, 1, 1)$, which has degree four, and relabel the remaining points in the inherited order, starting at zero. The resulting graph has 46 edges. Adding $\{0, 9\}$ gives exactly

$$(0, 1, 5, 9), \quad (0, 9, 18, 12), \quad (0, 9, 15, 16).$$

The full edge lists and exact counts are included in the certificates. Thus the upper bounds require only checking these finite graphs; no classification of polarity graphs is used.

Proof of Theorem 1. Lemma 2 proves the order-19 equality. Proposition 4, the order-20 witness, and $e_{20} = 46$ give $h(20) = 3$. If a 21-vertex, 51-edge graph had at most two cycles, it would have a vertex of degree at most $\lfloor 102/21 \rfloor = 4$. Deleting it leaves at least 47 edges on 20 vertices; retaining 47 edges contradicts Proposition 4. The order-21 witness and $e_{21} = 50$ give the matching upper bound. $\square$

5 Reproducibility and limitations

The accompanying `supplementary.zip` contains the exact graph inputs, the enumeration and verification programs, and their recorded outputs. Its README distinguishes generation, checking individual witnesses, and checking exhaustive coverage. The main replay program is `reproduce_candidate.py`. The README identifies the production enumerators and the separate attachment-block and catalogue verifiers. The replay uses Python 3.10 or later (the recorded completed run used Python 3.14.7) and nauty 2.9.3. The two `geng` commands above are the complete base-generation domains. Reproduction should run in an extracted working copy because the original programs write their output files. Recorded installation paths are documented in the README.

The external trust boundary consists of the attributed catalogue thresholds, the completeness of its five order-19 extremal classes, and nauty’s base generation. Separate exact implementations check the rooted attachment enumeration, extension exclusions, and witnesses. Their agreement is additional verification, not a formal proof-assistant certificate or a proof of the external catalogue. The construction and theorem give a finite benchmark for supersaturation searches. They do not determine $h(n)$ for $n \geq 22$, certify priority, or resolve the original asymptotic assertion.

Computational and AI assistance. OpenAI Codex assisted with the reductions, exact Python programs, source comparison, and preparation of this manuscript. A separate model context examined the frozen mathematical argument and produced the attachment-block and cycle-traversal verification programs. That review is not journal peer review or statistically independent evidence. All reported graph tests use exact integer and set operations; no floating-point fit, confidence estimate, or language-model judgment serves as a mathematical certificate.

References

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